Online Appendix of "Learning in a Small/Big World"

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March 21, 2024

For online publication only

A Switching between different automatons

This section illustrates how the setup in this paper encompasses learning mechanisms that involve switching between automatons with size smaller than M.

I first argue that assuming that the DM could switch between multiple learning mechanisms with M memory states implicitly implies that he has a larger memory capacity than M. Suppose that a DM starts with $(\mathscr{T}, d)$ at memory state m_1 and switches to $(\mathscr{T}', d')$ and $(\mathscr{T}'', d'')$ once he transit to memory state m_1 and m_2 respectively. As $(\mathscr{T}, d) \neq (\mathscr{T}', d') \neq (\mathscr{T}'', d'')$, when the DM receives a signal s at memory state m and decides to which memory state he transit to, or when he decides which action he takes at memory state m, he has to remember whether he has once transited to memory state m_1 and m_2. In other words, the DM has to track not only his current memory states, but also has to memorize the (incomplete) history of his previous memory states. Switching between multiple automatons thus implicitly implies a larger memory capacity.

Now, I illustrate through an example that a M memory states automaton could be designed to involve switching between automatons with smaller sizes. The example is illustrated in Figure A.1. In this example, the DM starts at memory state 3 with a learning mechanism $(\mathscr{T}, d)$ that involves 5 memory states 1 to 5. Once he transits to memory state 1, he switches to another learning mechanism $(\mathscr{T}', d')$ that involves 5 memory states 1, $10, 11, 12, 13$. On the other hand, once he transits to memory state 5, he switches to the learning mechanism $(\mathscr{T}', d')$ that involves 5 memory states $5, 6, 7, 8, 9$. Thus, the proposed 13 memory states automaton can be interpreted as a mechanism that involves switching between three 5 memory states automatons.

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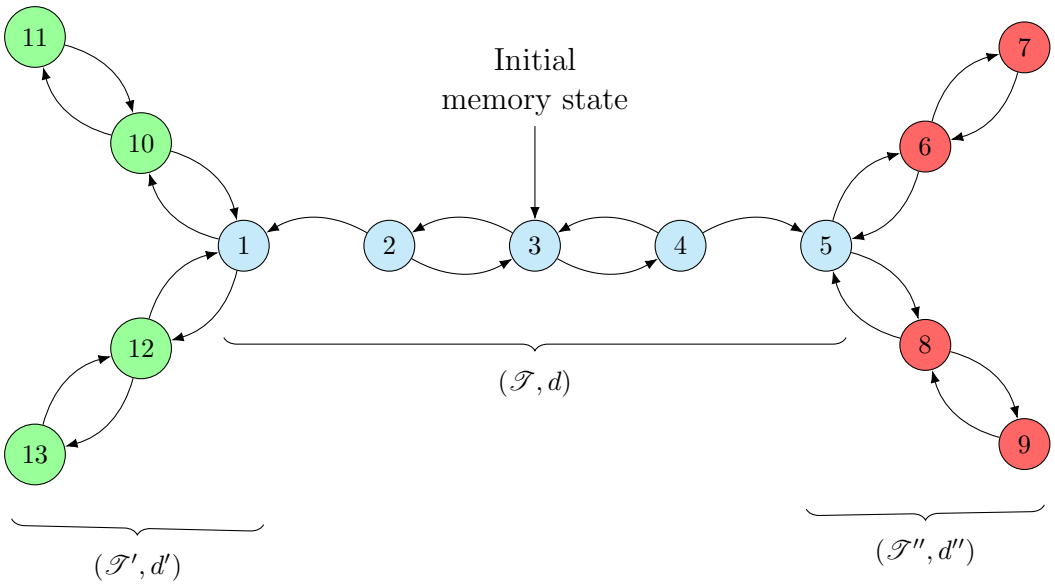


Figure A.1: An example of a 13 memory states automaton that involves switching between three 5 memory states automatons. The DM starts at memory state 3 with $(\mathscr{T}, d)$, and switches to $(\mathscr{T}', d')$ and $(\mathscr{T}'', d'')$ once he transits to memory state 1 and 5, respectively.

B Discounted future utility

In this section, I looked into the alternative formulation where future utility are discounted. More specifically, we look into the following updating mechanism

$$(m_1^{*D}(\rho), \mathscr{T}^{*D}(\rho), d^{*D}(\rho)) = \arg\max_{m_1, \mathscr{T}, d} E\left[(1-\rho)\sum_{t=1}^{\infty} \rho^t u(a_t, \omega)\right],$$

for some discount factor $\rho < 1$. We have the following result.

Proposition B.1. *For all $\epsilon > 0$, there exists some $\bar{\rho} < 1$ such that $L(m_1^{*D}(\rho), \mathscr{T}^{*D}(\rho), d^{*D}(\rho)) < L_M^* + \epsilon$ for all $\rho > \bar{\rho}$.*

In other words, for ρ close to 1, the updating mechanism that maximizes discounted sum of utility, $(m_1^{*D}(\rho), \mathscr{T}^{*D}(\rho), d^{*D}(\rho))$, is also ϵ-optimal for very small ϵ. Thus, the results in the main text continues to hold.

C Infinite N

In this section, I analyze an extension where $N, M \to \infty$ and characterize a sequence of updating mechanism $\mathscr{T}_{N,M}$ that is ϵ-optimal at the limit where $N, M \to \infty$. Moreover, I focus on cases

where $N = \Theta(M^h)$, i.e., N goes to infinite as fast as M^h. In this setting, small worlds refer to the case where $h < 1$ and big worlds refer to the case where $h \geq 1$. Note that when $h < 1$, as $N, M \to \infty$, $\frac{N}{M} \to 0$; when $h \geq 1$, $\lim_{N,M\to\infty} \frac{N}{M} > 0$. In the following, I show that the results in the baseline model hold qualitatively in this extension. It thus shows that the behavioral implications depend on both N and M, and more specifically the ratio $\frac{N}{M}$, instead of just the absolute value of M.

Formally, I consider a sequence of inference problems, which are characterized by the utilities, prior beliefs and information structures, i.e., $\{(\boldsymbol{u}_N^\omega, p_N^\omega, f_N^\omega)_{\omega=1}^N\}_{N=1}^\infty$. The subscript N emphasizes the fact that the utilities, prior beliefs, and information structures could change along the sequence. A small world is a sub-sequence of inference problems with a sequence of memory size M, and is denoted by $\left\{(\boldsymbol{u}_{N(k)}^\omega, p_{N(k)}^\omega, f_{N(k)}^\omega)_{\omega=1}^{N(k)}, M(k)\right\}_{k=1}^\infty$ where $\lim_{k\to\infty} N(k), M(k) \to \infty$ and $N(k) = \Theta(M(k)^h)$ with $h < 1$. For example, $N(k) = k$ and $M(k) = k^2$, such that the sequence $M(k)$ dominates the sequence $N(k)$ at the limit. A big world is defined analogously with $h \geq 1$. For ease of exposition, I sometimes omit the argument k in the sequences $N(k)$ and $M(k)$.

Before I present the result, it is necessary to present some additional assumptions about the nature of the sequence of inference problems. First, I assume that $u_N(\omega, \omega) - \max_{\omega' \neq \omega} u_N(\omega', \omega) \in [\underline{u}, \overline{u}]$ for some $\underline{u} > 0$ for all ω and all N. Thus along the sequence of inference problems, no states are infinitely more important than another state. The prior belief of the DM satisfies the following full support assumption:

Assumption D.1. *For any sequence of subset of states $\Omega_N \subseteq \{1, \cdots, N\}$ with cardinality $|\Omega_N|$ and a well-defined limit $\lim_{N\to\infty} \frac{|\Omega_N|}{N}$,*

$$\lim_{N\to\infty} \sum_{\omega\in\Omega_N} p_N^\omega > 0 \text{ if } \lim_{N\to\infty} \frac{|\Omega_N|}{N} > 0. \tag{C.1}$$

That is, Equation (C.1) ensures any sequence of subsets of states that has a non-negligible measure in fractions at the limit, i.e., $\lim_{N\to\infty} \frac{|\Omega_N|}{N} > 0$, also has a non-negligible probability mass at the limit, i.e., $\lim_{N\to\infty} \sum_{\omega\in\Omega_N} p_N^\omega > 0$.[1] This is analogue to the full support assumption in the baseline model. A simple example would be $p_N^\omega = \frac{1}{N}$ for all ω and N.

Similar to the baseline model, I assume that no signals rule out any states of the world:

[1] For example, it rules out prior beliefs $(p_N^\omega)_{\omega=1}^N$ where $p^\omega = \frac{1}{N^2}$ for $\omega \leq \frac{N}{2}$ and $p^\omega = \frac{2N-1}{N^2}$ for $\omega > \frac{N}{2}$, i.e., first half of all possible states have negligible probability mass at the limit.

there exists $\varsigma > 0$ such that

$$\inf_{s \in S} \frac{f_N^{\omega}(s)}{f_N^{\omega'}(s)} > \varsigma \text{ for all } \omega, \omega' \in \Omega \text{ and for all } N. \tag{C.2}$$

Finally, I make the following assumption about identifiability that plays similar role to the assumption that $f^{\omega} \neq f^{\omega'}$ in the baseline model. It ensures that signal structures with negligible Cauchy-Schwarz distances must have negligible probability mass at the limit.

Assumption D.2. *For all $\varepsilon > 0$, there exists some $\xi > 0$ and some sequence of subsets of states $N_\xi \subseteq \{1, \cdots, N\}$ such that $\lim_{N\to\infty} \left(\sum_{\omega \in N_\xi} p_N^{\omega}\right) > 1 - \varepsilon$ and*

$$\lim_{N\to\infty} \inf_{\omega, \omega' \in N_\xi; \omega' \neq \omega} \left\{ -\log \frac{\int f_N^{\omega}(s) f_N^{\omega'}(s) ds}{\sqrt{\int (f_N^{\omega}(s))^2 ds} \sqrt{\int (f_N^{\omega'}(s))^2 ds}} \right\} > \xi,$$

Appendix D provides an example of the signal structures that satisfy Assumption D.2 and Equation C.2. Analogue to the main text, L_k^* denotes the infinmum of the asymptotic utility loss with $N(k)$ and $M(k)$, $\lim_{k\to\infty} L_k^*$ denotes the infinmum of the asymptotic utility as k goes to infinite, and a sequence of updating mechanism $\{(m_{1k}, \mathscr{T}_k, d_k)\}_{k=1}^{\infty}$ is ϵ-optimal (at the limit) if $\lim_{k\to\infty} L_k(m_{1k}, \mathscr{T}_k, d_k) \leq \lim_{k\to\infty} L_k^* + \epsilon$.

C.1 Small Worlds

First, I present the results in small worlds. The following proposition shows that actions taken by the DM is close to the Bayesian benchmark.

Proposition C.1. *Fix $\left\{(\boldsymbol{u}_{N(k)}^{\omega}, p_{N(k)}^{\omega}, f_{N(k)}^{\omega})_{\omega=1}^{N(k)}, M(k)\right\}_{k=1}^{\infty}$ where $N(k) = \Theta(M_i(k)^{h_i})$ with $h < 1$. For all $\epsilon > 0$, there exists a sequence of updating mechanism $\{m_{1k}, \mathscr{T}_k, d_k\}_{k=1}^{\infty}$ such that $\lim_{k\to\infty} L(m_{1k}, \mathscr{T}_k, d_k) < \epsilon$. Therefore, $\lim_{k\to\infty} L_k^* = 0$*

The intuition and the proof of the proposition follow similar arguments in Proposition 1. In particular, I show that the simple updating mechanism illustrated in Figure 2 achieves perfect asymptotic learning. Similar to the baseline model, as $\frac{M}{N} \to \infty$, the DM could allocate infinite memory states to each state, and his confidence level for each action ranges from 1 to ∞. As the DM could be almost sure about taking each action, he makes no mistakes asymptotically and he almost always matches his action with the true state.

Next, as the DM can learn perfectly under every state of the world, he has no incentive to ignore some states of the world to focus on a strict subset of states, i.e., ignorance in learning

is not optimal.

Corollary C.1. *Fix* $\left\{(\boldsymbol{u}^{\omega}_{N(k)}, p^{\omega}_{N(k)}, f^{\omega}_{N(k)})_{\omega=1}^{N(k)}, M(k)\right\}_{k=1}^{\infty}$ *where* $N(k) = \Theta(M(k)^h)$ *with* $h < 1$. *A sequence of updating mechanisms* $\{m_{1k}, \mathscr{T}_k, d_k\}_{k=1}^{\infty}$ *is* ϵ*-optimal at the limit* $k \to \infty$ *only if it ignores at most* $\frac{\epsilon}{\underline{u}}$ *measures of states at the limit.*

In particular, when $\epsilon \to 0$, Corollary C.1 shows that the optimal updating mechanism must ignore at most a negligible amount of states. To complete the analysis in small worlds, the following corollary shows that different individuals are bound to agree in small worlds if they agree on the probability 0 events.[2]

Corollary C.2. *Fix* $\left\{(\boldsymbol{u}^{\omega}_{N(k)i}, p^{\omega}_{N(k)i}, f^{\omega}_{N(k)i})_{\omega=1}^{N(k)}, M_i(k)\right\}_{k=1}^{\infty}$ *where* $N(k) = \Theta(M_i(k)^{h_i})$ *with* $h_i < 1$ *for* $i = A, B$. *If* $\lim_{N\to\infty} p^{\omega}_{NA} > 0$ *if and only if* $\lim_{N\to\infty} p^{\omega}_{NB} > 0$ *for all* $\omega \in \{1, \cdots \infty\}$, *individual A and B must almost always agree with each other at the limit if they adopt a sequence of* ϵ*-optimal updating mechanism with* $\epsilon \to 0$.

C.2 Big Worlds

Now I present the results in big worlds. First, the following Proposition shows that asymptotic learning is different than Bayesian.

Proposition C.2. *Fix* $\left\{(\boldsymbol{u}^{\omega}_{N(k)}, p^{\omega}_{N(k)}, f^{\omega}_{N(k)})_{\omega=1}^{N(k)}, M(k)\right\}_{k=1}^{\infty}$ *where* $N(k) = \Theta(M(k)^h)$ *with* $h \geq 1$, $\lim_{k\to\infty} L^*_k > 0$.

The intuition is again similar to Proposition 1 in the baseline model. In this extension, similar to the baseline model, $\frac{M}{N} < \infty$ implies that the DM can only allocate a finite number of memory states to almost all states of the world. However, to make no mistakes when choosing action ω, the DM has to record infinite signals supporting state ω and be almost sure about state ω. As $\frac{M}{N} < \infty$ is bounded, the DM cannot be allocate infinite number of memory states to record supporting information for each state of the world. Thus he cannot be almost sure for almost all actions and is bound to make mistakes. Next, I show the analogue of Proposition 2 about ignorance in this extension.

Proposition C.3. *Fix* $\left\{(\boldsymbol{u}^{\omega}_{N(k)}, p^{\omega}_{N(k)}, f^{\omega}_{N(k)})_{\omega=1}^{N(k)}, M(k)\right\}_{k=1}^{\infty}$ *where* $N(k) = \Theta(M(k)^h)$ *with* $h \geq 1$. *All updating mechanisms must ignore almost all (measured in fraction) states at the limit.*

[2]Note that in the baseline model, individuals necessarily agree on the probability 0 events as $p^{\omega} > 0$ for all ω. In contrast, Assumption D.1 does not guarantee an agreement: there may exist some ω such that $\lim_{N\to\infty} p^{\omega}_{NA} > 0$ but $\lim_{N\to\infty} p^{\omega}_{NB} = 0$.

Note that the statements in Proposition 2 and C.3 are different. Proposition 2 indicates that there exist some prior beliefs such that all (almost) optimal updating mechanisms will ignore some states. In contrast, Proposition C.3 is "stronger": for all prior beliefs, all learning mechanisms, including the (almost) optimal updating mechanism, have to be ignorant, and has to ignore almost all states. Roughly speaking, the intuition of Proposition C.3 is as follows: when $M \to \infty$, the stationary probability of each memory state becomes infinitesimally small. As the DM cannot allocate infinite memory state in almost all M^ω, i.e., the DM takes almost all action ω in a finite number of memory states, he must choose almost all actions with 0 probability.

To complete the analysis in the extension, the following corollary shows that disagreement could be persistent in big worlds, as in the baseline model.

Corollary C.3. *Fix $N(k)$ and $M(k)$ such that $N(k) = \Theta(M_i(k)^{h_i})$ with $h_i < 1$ for $i = A, B$. Also assume that $\lim_{N\to\infty} p^\omega_{NA} > 0$ if and only if $\lim_{N\to\infty} p^\omega_{NB} > 0$ for all $\omega \in \{1, \cdots \infty\}$. There exists some $\{(\boldsymbol{u}^\omega_{N(k)i}, p^\omega_{N(k)i}, f^\omega_{N(k)i})^N_{\omega=1}\}^\infty_{k=1}$ where $i = A, B$, such that for all ϵ, there exists some sequences of ϵ-optimal updating mechanism for individual A and B such that they almost always disagree.*

Note that Corollary C.3 is weaker than Corollary 2 in the sense that there exist some, but not all, ϵ-optimal updating mechanisms that lead to disagreement. The intuition is as follows: as shown in Proposition C.3, the DM must ignore almost all actions. Thus, he obtains positive utility only under a negligible amount of states, and it implies that a large set of updating mechanisms could be ϵ-optimal. In particular, when the prior likelihood of all states are infinitesimally small at the limit, e.g., $p^\omega = \frac{1}{N}$, the supremum utility equals 0 and all updating mechanisms are ϵ-optimal. This leaves ample room for the individuals to adopt different optimal learning mechanisms and thus leads to disagreement. By comparing Corollary C.3 and C.2, we can conclude that the same result in the baseline model holds qualitatively in this extension: asymptotic disagreement does not arise in small worlds but could arise (optimally) in big worlds.

D Example of a sequence of signal structures that satisfies Equation C.2 and Assumption D.2

In this section, I provide an example of a sequence of signal structures that satisfies Equation C.2 and Assumption D.2. Consider the following seqeunce of signal structures $\{(f^\omega_N)^N_{\omega=1}\}^\infty_{N=1}$ with

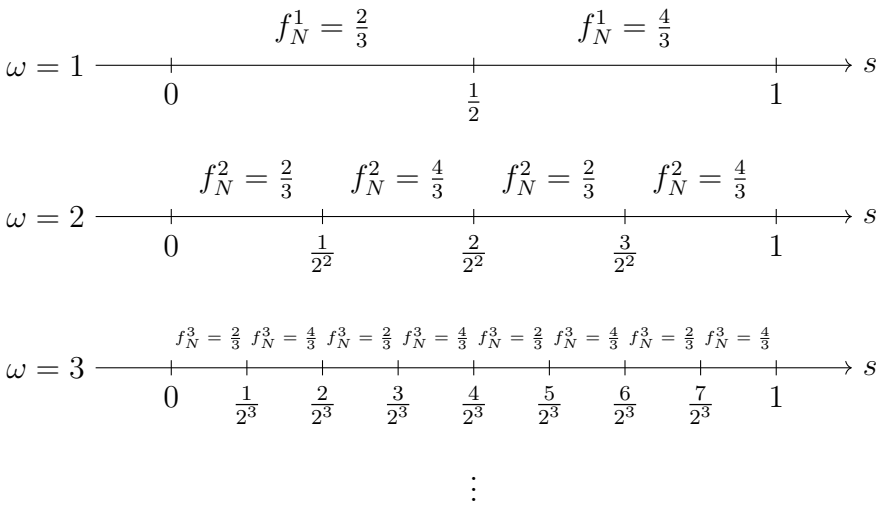


Figure D.2: Signal structures that satisfies Equation C.2 and Assumption D.2 as $N \to \infty$. The signal structures comprise low and high density alternatively in 2^ω equal-sized intervals.

$S = [0, 1)$:

$$f_N^\omega(s) = \begin{cases} \frac{2}{3} \text{ for } s \in [\frac{i}{2^\omega}, \frac{i+1}{2^\omega}) \text{ where } i = 0, 2, \cdots, 2^\omega - 2; \\ \frac{4}{3} \text{ for } s \in [\frac{i}{2^\omega}, \frac{i+1}{2^\omega}) \text{ where } i = 1, 3, \cdots, 2^\omega - 1. \end{cases}$$

The signal structures are illustrated in Figure D.2. First, they satisfy Equation C.2 as $\frac{f_N^\omega(s)}{f_N^{\omega'}(s)} \geq \frac{1}{2}$ for all $\omega, \omega' \in \Omega$, N and s. Second, for any N and given any ω and $\omega' \neq \omega$, the Cauchy-Schwarz distance is equal to:

$$\begin{aligned} -\log \frac{\int f_N^\omega(s) f_N^{\omega'}(s) ds}{\sqrt{\int (f_N^\omega(s))^2 ds}\sqrt{\int (f_N^{\omega'}(s))^2 ds}} &= -\log \frac{\frac{1}{4}(\frac{2}{3})^2 + \frac{1}{2}\frac{2}{3}\frac{4}{3} + \frac{1}{4}(\frac{4}{3})^2}{\frac{1}{2}(\frac{2}{3})^2 + \frac{1}{2}(\frac{4}{3})^2} \\ &= -\log \frac{1}{10/9} > 1 + \xi \end{aligned}$$

where $\xi < \log \frac{10}{9} - 1$ and thus satisfies Assumption D.2.

E Robustness of the results in small worlds to updating mistakes

Below, I show that the behavioral implications is small world, i.e., asymptotic behavior is close to Bayesian, and that disagreement does not persist, hold even when individuals make "updating mistakes". In other words, the results are robust to individuals' limited ability to

design and follow an "optimal" updating mechanism.

Consider two individuals A and B. Individual A adopts the simple updating mechanism described in the proof of Proposition 1 while individual B "attempts" to adopt the same updating mechanism but makes local mistakes as he randomly transits to neighboring memory states with some probability $\gamma \in (0,1)$. Formally, the transition rule of individual B, denoted as $\mathscr{T}'(m,s)$, is as follows:

$$\mathscr{T}'(0,s) = \begin{cases} \mathscr{T}(0,s) & \text{with probability } 1-\gamma; \\ ji & \text{with probability } \frac{\gamma}{N} \text{ for all } j = 1, \cdots, N. \end{cases}$$

$$\mathscr{T}'(i1,s) = \begin{cases} \mathscr{T}(i1,s) & \text{with probability } 1-\gamma; \\ i2 & \text{with probability } \frac{\gamma}{2}; \\ 0 & \text{with probability } \frac{\gamma}{2}. \end{cases}$$

$$\mathscr{T}'(i\lambda,s) = \begin{cases} \mathscr{T}(i\lambda,s) & \text{with probability } 1-\gamma; \\ i(\lambda-1) & \text{with probability } \gamma. \end{cases}$$

while for $k = 2, 3, \cdots, \lambda - 1$,

$$\mathscr{T}'(ik,s) = \begin{cases} \mathscr{T}(ik,s) & \text{with probability } 1-\gamma; \\ i(k-1) & \text{with probability } \frac{\gamma}{2}; \\ i(k+1) & \text{with probability } \frac{\gamma}{2}. \end{cases}$$

where $\mathscr{T}(m,s)$ is defined in the proof of Proposition 1. The initial memory states and decision rule of individual B stay the same as individual A. Such updating mistakes could be induced by memory imperfections, i.e., the DM's memory state is subject to local fluctuations, or imperfect perceptions on signals, e.g., the DM may mistakenly perceive any signal as a signal that supports state ω.

Proposition E.1. *Consider individual A who adopts the updating mechanism $\mathscr{T}$ and individual B who makes local mistakes with some probability $\gamma \in (0,1)$, characterized above by $\mathscr{T}'(m,s)$. Fix N and $(\boldsymbol{u}_B^\omega, p_B^\omega, f_B^\omega)_{\omega=1}^N$, for all $\gamma \in (0,1)$,*

$$\lim_{k\to\infty} L_k(m_1, \mathscr{T}', d) = 0.$$

Moreover, individual A and B are bound to agree in small worlds.

F Proof of the results in online appendix

F.1 Proof of Proposition B.1

Proof. Suppose to the contrary that $\lim_{\rho\to 1} L(m_1^{*D}(\rho), \mathscr{T}^{*D}(\rho), d^{*D}(\rho)) > L_M^* + \epsilon$ for some $\epsilon > 0$. It implies that there exists some updating mechanism $(m_1, \mathscr{T}, d)$ that yields strictly higher asymptotic utility than $(m_1^{*D}(\rho), \mathscr{T}^{*D}(\rho), d^{*D}(\rho))$ for all ρ close to 1. By continuity, $(m_1, \mathscr{T}, d)$ also yields strictly higher discounted sum of utility than $(m_1^{*D}(\rho), \mathscr{T}^{*D}(\rho), d^{*D}(\rho))$ for all ρ close to 1. The result follows by contradiction. □

F.2 Proof of Proposition C.1

Proof. I show that for all $\epsilon > 0$, there exists a sequence of updating mechanisms with utility losses that converge to smaller than ϵ as $k \to \infty$. First, by Assumption D.2, for all $\frac{\epsilon}{\bar{u}} > 0$, there exists an $\xi > 0$ and a sequence of subset of states N_ξ such that $\lim_{N\to\infty}\left(\sum_{\omega\in N_\xi} p_N^\omega\right) > 1 - \frac{\epsilon}{\bar{u}}$ and

$$\lim_{N\to\infty} \inf_{\omega,\omega'\in N_\xi;\omega'\neq\omega} \left\{ -\log \frac{\int f_N^\omega(s) f_N^{\omega'}(s) ds}{\sqrt{\int (f_N^\omega(s))^2 ds}\sqrt{\int (f_N^{\omega'}(s))^2 ds}} \right\} > \xi$$

Before proving the Proposition, I first prove the following Lemma for $(G_N^\omega(s))_{\omega=1}^N$ defined as:

$$G_N^\omega(s) = \frac{f_N^\omega(s)}{\sqrt{\int (f_N^\omega(s))^2 ds}}.$$

Lemma F.1. *For $\frac{\epsilon}{\bar{u}} > 0$, there exists an $\tilde{\xi} > 0$, a sequence of $\delta_N > 1$ and a sequence of subset of states N_ξ with $\lim_{N\to\infty}\left(\sum_{\omega\in N_\xi} p_N^\omega\right) > 1 - \frac{\epsilon}{\bar{u}}$ such that*

$$\begin{aligned} &\frac{\delta_N F_N^\omega(G_N^{\omega'})}{\sum_{\omega''\neq\omega'} F_N^\omega(G_N^{\omega''})} > 2 \text{ for all } \omega,\omega' \in \Omega \text{ and } N; \\ &\lim_{N\to\infty} \inf_{\omega,\omega'\in N_\xi;\omega'\neq\omega} \frac{\frac{\delta_N F_N^\omega(G_N^\omega)}{\sum_{\omega''\neq\omega} F_N^\omega(G_N^{\omega''})}}{\frac{\delta_N F_N^\omega(G_N^{\omega'})}{\sum_{\omega''\neq\omega'} F_N^\omega(G_N^{\omega''})}} > 1 + \tilde{\xi}. \end{aligned} \tag{F.1}$$

Proof. As $F_N^\omega(G_N^{\omega'}) > 0$ for all ω, ω', there always exists a big enough δ_N such that the first

inequality of Equation (F.1) holds. To prove the second Inequality, note that

$$\begin{aligned}
&\lim_{N\to\infty}\inf_{\omega,\omega'\in N_\xi;\omega'\neq\omega}\frac{\frac{\delta_N F_N^\omega(G_N^\omega)}{\sum_{\omega''\neq\omega}F_N^\omega(G_N^{\omega''})}}{\frac{\delta_N F_N^\omega(G_N^{\omega'})}{\sum_{\omega''\neq\omega'}F_N^\omega(G_N^{\omega''})}}\\
\geq&\lim_{N\to\infty}\inf_{\omega,\omega'\in N_\xi;\omega'\neq\omega}\frac{F_N^\omega(G_N^\omega)}{F_N^\omega(G_N^{\omega'})} \qquad \text{(F.2)}\\
=&\lim_{N\to\infty}\inf_{\omega,\omega'\in N_\xi;\omega'\neq\omega}\frac{\sqrt{\int(f_N^\omega(s))^2ds}\sqrt{\int(f_N^{\omega'}(s))^2ds}}{\int f_N^\omega(s)f_N^{\omega'}(s)ds}\\
>&\exp(\xi)\\
=&1+\tilde{\xi}
\end{aligned}$$

where $\tilde{\xi}=1-\exp(\xi)$ and the first inequality of Equation (F.2) is implied by the fact that $F_N^\omega(G_N^\omega)\geq F_N^\omega(G_N^{\omega'})$ and thus $\frac{\sum_{\omega''\neq\omega'}F_N^\omega(G_N^{\omega''})}{\sum_{\omega''\neq\omega}F_N^\omega(G_N^{\omega''})}\geq 1$. □

Now consider a sequence of updating mechanisms described in the proof of Proposition 1, but include only states in N_ξ, i.e., there are only branches that correspond to actions in N_ξ, and there exists no $m\in M$ such that $d(m)=\omega'$ for $\omega'\notin N_\xi$. As in the proof of Proposition 1, denote $\frac{\delta_N F_N^\omega(G_N^{\omega'})}{\sum_{\omega''\neq\omega'}F_N^\omega(G_N^{\omega''})}$ as $r_N^{\omega\omega'}$, the stationary distribution under state $\omega\in N_\xi$ is as follows:

$$\mu_{N\omega\lambda}^\omega\sum_{i=1}^{\lambda}\left[r_N^{\omega\omega}\right]^{-(\lambda-i)}+\mu_{N\omega\lambda}^\omega\left[r_N^{\omega\omega}\right]^{-\lambda}+\mu_{N\omega\lambda}^\omega\left[r_N^{\omega\omega}\right]^{-\lambda}\sum_{\omega'\in N_\xi\setminus\{\omega\}}\sum_{i=1}^{\lambda}\left[r_N^{\omega\omega'}\right]^{(\lambda-i)}=1 \qquad \text{(F.3)}$$

where μ_{Nm}^ω is the stationary probability in memory state m under state ω when state space has size N. Given the first inequality of Lemma F.1, $r_N^{\omega\omega}>2$ for all ω and N and therefore $\lim_{k\to\infty}\mu_{N(k)\omega\lambda}^\omega\left[r_{N(k)}^{\omega\omega}\right]^{-\lambda(k)}=0$. On the other hand,

$$\begin{aligned}
\left[r_N^{\omega\omega}\right]^{-\lambda}\sum_{\omega'\in N_\xi\setminus\{\omega\}}\sum_{i=1}^{\lambda}\left[r_N^{\omega\omega'}\right]^{(\lambda-i)}&=\left[r_N^{\omega\omega}\right]^{-\lambda}\sum_{\omega'\in N_\xi\setminus\{\omega\}}\frac{\left[r_N^{\omega\omega'}\right]^{\lambda}-1}{r_N^{\omega\omega'}-1}\\
&\leq\sum_{\omega'\in N_\xi\setminus\{\omega\}}\left[\left[\frac{r_N^{\omega\omega'}}{r_N^{\omega\omega}}\right]^{\lambda}-\left[r_N^{\omega\omega'}\right]^{-\lambda}\right]\\
&\leq\sum_{\omega'\in N_\xi\setminus\{\omega\}}(1+\tilde{\xi})^{-\lambda}\leq N(1+\tilde{\xi})^{-\lambda}
\end{aligned}$$

where the first inequality is implied by the first inequality in Lemma F.1, and the second inequality is implied by the second inequality in Lemma F.1 and that $\left[r_N^{\omega\omega'}\right]^{-\lambda} > 0$. As $(1+\tilde{\xi})^{-\lambda}$ converges to 0 exponentially and N converges to infinity linearly, $\lim_{k\to\infty} N(k)(1+\tilde{\xi})^{-\lambda(k)} = 0$. To see it formally, note that $N(k) = \Theta(M(k)^h)$ with $h < 1$ and $\lambda(k) = \frac{M(k)}{N(k)}$ imply that $\lambda(k) = \Theta(M(k)^{1-h})$. By the definition of Θ, we have for some constant K and some q big enough such that

$$\begin{aligned}
\frac{N(k)}{(1+\tilde{\xi})^{\lambda(k)}} \leq K \frac{M(k)^h}{(1+\tilde{\xi})^{M(k)(1-h)}} &= K\left[\frac{M(k)}{(1+\tilde{\xi})^{M(k)\frac{1-h}{h}}}\right]^h \\
&= K\left[\frac{M(k)}{(1+\tilde{\tilde{\xi}})^{M(k)}}\right]^h \\
&< K\left[\frac{M(k)}{\binom{M(k)}{2}\tilde{\tilde{\xi}}^2}\right]^h \\
&= K\left[\frac{2}{(M(k)-1)\tilde{\tilde{\xi}}^2}\right]^h
\end{aligned}$$

where $1+\tilde{\tilde{\xi}} = (1+\tilde{\xi})^{\frac{1-h}{h}}$ and the second inequality is implied by the Binomial Theorem. Thus,

$$\begin{aligned}
\lim_{k\to\infty}\left[r_N^{\omega\omega}\right]^{-\lambda} \sum_{\omega'\in N_\xi\setminus\{\omega\}} \sum_{i=1}^{\lambda}\left[r_N^{\omega\omega'}\right]^{(\lambda-i)} &\leq \lim_{k\to\infty}\left[r_N^{\omega\omega}\right]^{-\lambda} N(1+\tilde{\xi})^{-\lambda(k)} \\
&< \lim_{k\to\infty}\left[r_N^{\omega\omega}\right]^{-\lambda} K\left[\frac{2}{(M(k)-1)\tilde{\tilde{\xi}}^2}\right]^h = 0.
\end{aligned}$$

It implies that $\lim_{k\to\infty}\left[r_N^{\omega\omega}\right]^{-\lambda}\sum_{\omega'\in N_\xi\setminus\{\omega\}}\sum_{i=1}^{\lambda}\left[r_N^{\omega\omega'}\right]^{(\lambda-i)} = 0$ and

$$\lim_{k\to\infty}\sum_{i=1}^{\lambda}\mu_{N\omega i}^{\omega} = \lim_{k\to\infty}\mu_{N\omega\lambda}^{\mu}\sum_{i=1}^{\lambda}\left[r_N^{\omega\omega}\right]^{-(\lambda-i)} = 1 \tag{F.4}$$

for all $\omega \in N_\xi$. In state $\omega \in N_\xi$, the DM chooses action ω with asymptotic probability equal to 1. As $\lim_{N\to\infty}\sum_{\omega\in N_\xi} p_N^{\omega} > 1 - \frac{\epsilon}{\bar{u}}$, the asymptotic utility loss of the proposed non-ignorant updating mechanism is strictly bounded above by $\bar{u} \times \frac{\epsilon}{\bar{u}} = \epsilon$. □

F.3 Proof of Corollary C.1

Proof. Suppose in contrary the sequence of updating mechanisms ignore strictly more than $\frac{\epsilon}{\underline{u}}$ measures of states at the limit. That is, denoted $\tilde{N}(k)$ as the set of states that is ignored for a given sequence of updating mechanism at the limit, we have $\lim_{T\to\infty} E_{\omega'}\left[\frac{\sum_{t=1}^{T}\mathbb{1}_{a_t\in\tilde{N}(k)}}{T}\right]=0$ for (almost) all ω' and $\lim_{k\to\infty}\sum_{\omega\in\tilde{N}(k)} p^{\omega}_{N(k)} > \frac{\epsilon}{\underline{u}}$. The utility loss as $k\to\infty$ is

$$\lim_{k\to\infty} L_k \geq \lim_{k\to\infty}\sum_{\omega\in\tilde{N}(k)}\left[u_{N(k)}(\omega,\omega)-\max_{\omega'\neq\omega}u(\omega',\omega)\right]p^{\omega}_{N(k)}$$
$$\geq \underline{u}\lim_{k\to\infty}\sum_{\omega\in\tilde{N}(k)} p^{\omega}_{N(k)} > \underline{u}\frac{\epsilon}{\underline{u}}=\epsilon=\lim_{k\to\infty}L^{*}_{k}+\epsilon$$

which proves the result. □

F.4 Proof of Corollary C.2

Proof. With some abuse of notation, denote $\boldsymbol{\mathscr{T}^{\epsilon}_{ki}}$ as the set of ϵ-optimal updating mechanism of individual $i=A,B$. By Proposition C.1, we have:

$$\lim_{k\to\infty}\lim_{\epsilon\to 0}\sup_{(m_{1ki},\mathscr{T}_{ki},d_{ki})\in\boldsymbol{\mathscr{T}^{\epsilon}_{ki}}}\sum_{\omega}\left[\mathbb{1}\left\{\lim_{T\to\infty}E_{\omega,\mathscr{T}_{ki}}\frac{\sum_{t=1}^{T}\mathbb{1}_{a^i_t=\omega}}{T}=1\right\}p^{\omega}_{Ni}\right]=1$$

which is equivalent to

$$\lim_{k\to\infty}\lim_{\epsilon\to 0}\sup_{(m_{1ki},\mathscr{T}_{ki},d_{ki})\in\boldsymbol{\mathscr{T}^{\epsilon}_{ki}}}\sum_{\omega}\left[\mathbb{1}\left\{\lim_{T\to\infty}E_{\omega,\mathscr{T}_{ki}}\frac{\sum_{t=1}^{T}\mathbb{1}_{a^i_t=\omega}}{T}<1\right\}p^{\omega}_{Ni}\right]=0,$$

i.e., individual i would only take sub-optimal actions in probability 0 events measured by $(p^{\omega}_{Ni})^{N}_{\omega=1}$. As $\lim_{N\to\infty}p^{\omega}_{NA}>0$ if and only if $\lim_{N\to\infty}p^{\omega}_{NB}>0$ for all ω, combined with Assumption D.1, this implies that

$$\lim_{k\to\infty}\sum_{\omega\in\tilde{\Omega}(k)}p^{\omega}_{NA}=0 \text{ if and only if } \lim_{k\to\infty}\sum_{\omega\in\tilde{\Omega}(k)}p^{\omega}_{NB}=0$$

for all $\tilde{\Omega}(k)\in\{1,\cdots,N(k)\}$. Thus,

$$\lim_{k\to\infty}\lim_{\epsilon\to 0}\sup_{(m_{1ki},\mathscr{T}_{ki},d_{ki})\in\boldsymbol{\mathscr{T}^{\epsilon}_{ki}}}\sum_{\omega}\left[\mathbb{1}\left\{\lim_{T\to\infty}E_{\omega,\mathscr{T}_{ki}}\frac{\sum_{t=1}^{T}\mathbb{1}_{a^i_t=\omega}}{T}<1\right\}p^{\omega}_{NI}\right]=0 \text{ for all } i,I\in\{A,B\}$$

and

$$\lim_{k\to\infty}\lim_{\epsilon\to 0}\sup_{(m_{1kA},\mathscr{T}_{kA},d_{kA})\in\boldsymbol{\mathscr{T}^{\epsilon}_{kA}},(m_{1kB},\mathscr{T}_{kB},d_{kB})\in\boldsymbol{\mathscr{T}^{\epsilon}_{kB}}}\sum_{\omega}\left[\mathbb{1}\left\{\lim_{T\to\infty}E_{\omega,\mathscr{T}_{kA},\mathscr{T}_{kB}}\frac{\sum_{t=1}^{T}\mathbb{1}_{a_t^A=a_t^B=\omega}}{T}<1\right\}p^{\omega}_{Ni}\right]$$
$$\leq\lim_{k\to\infty}\lim_{\epsilon\to 0}\sup_{(m_{1kA},\mathscr{T}_{kA},d_{kA})\in\boldsymbol{\mathscr{T}^{\epsilon}_{kA}}}\sum_{\omega}\left[\mathbb{1}\left\{\lim_{T\to\infty}E_{\omega,\mathscr{T}_{kA}}\frac{\sum_{t=1}^{T}\mathbb{1}_{a_t^A=\omega}}{T}<1\right\}p^{\omega}_{Ni}\right]$$
$$+\lim_{k\to\infty}\lim_{\epsilon\to 0}\sup_{(m_{1kB},\mathscr{T}_{kB},d_{kB})\in\boldsymbol{\mathscr{T}^{\epsilon}_{kB}}}\sum_{\omega}\left[\mathbb{1}\left\{\lim_{T\to\infty}E_{\omega,\mathscr{T}_{kB}}\frac{\sum_{t=1}^{T}\mathbb{1}_{a_t^B=\omega}}{T}<1\right\}p^{\omega}_{Ni}\right]$$
$$=0$$

for $i=A,B$. The result follows. □

F.5 Proof of Proposition C.2

Proof. First, note that for $\lim_{k\to\infty}L^*_k=0$, denote $M^{\omega}_k=\{m:d_k(m)=\omega\}$, we must have for some $\{m_{1k},\mathscr{T}_k,d_k\}_{k=1}^{\infty}$,

$$\lim_{k\to\infty}\sum_{m\in M^{\omega}_k}\mu^{\omega}_{N(k)m}=1$$

for almost all ω, i.e., there must exist a sequence of subsets of states $\hat{\Omega}(k)\in\{1,\cdots,N(k)\}$ where $\lim_{k\to\infty}\sum_{\omega\in\hat{\Omega}(k)}p^{\omega}_{N(k)}=1$ and

$$\lim_{k\to\infty}\sum_{m\in M^{\omega}_k}\mu^{\omega}_{N(k)m}=1.$$

for all $\omega\in\hat{\Omega}(k)$. Additionally, Assumption D.1 implies that there must exist a sequence of subsets of states $\hat{\Omega}(k)\in\{1,\cdots,N(k)\}$ where $\lim_{k\to}\frac{|\hat{\Omega}(k)|}{N(k)}=1$ and

$$\lim_{k\to\infty}\sum_{m\in M^{\omega}_k}\mu^{\omega}_{N(k)m}=1,$$

for all $\omega\in\hat{\Omega}(k)$. That is, the DM chooses the optimal action in almost all states, measured in both prior probability or in fraction. It implies that for all ω in $\hat{\Omega}(k)$, there must exist a set of

memory state $\hat{M}_k^\omega \subseteq M_k^\omega$ such that

$$\begin{aligned}
&\lim_{k\to\infty} \sum_{m\in\hat{M}_k^\omega} \mu^\omega_{N(k)m} = 1; \\
&\lim_{k\to\infty} \sum_{m\in\hat{M}_k^\omega} \mu^{\omega'}_{N(k)m} = 0 \text{ for all } \omega' \in \hat{\Omega}(k)\setminus\{\omega\}; \\
&\lim_{k\to\infty} \max_{m\in\hat{M}_k^\omega} \frac{\mu^\omega_{N(k)m}}{\mu^{\omega'}_{N(k)m}} = \infty \text{ for all } \omega' \in \hat{\Omega}(k)\setminus\{\omega\}.
\end{aligned} \tag{F.5}$$

In the following I prove that for Equation (F.5) to hold, $\frac{M}{N}$ has to go to ∞. First consider an irreducible automaton $\mathscr{T}_k$. Fix a $\omega' \in \hat{\Omega}(k)\setminus\{\omega\}$, without loss of generality rearrange the memory states such that $\frac{\mu^\omega_{N(k)m}}{\mu^{\omega'}_{N(k)m}}$ is weakly decreasing in m. As argued in the proof of Proposition 2, Lemma 2 of Hellman and Cover (1970) holds in the current setting, so that for all $m < M(k)$,

$$\begin{aligned}
\frac{\mu^\omega_{N(m+1)}}{\mu^{\omega'}_{N(m+1)}} &\geq (\bar{l}_N^{\omega\omega'}\bar{l}_N^{\omega'\omega})^{-1}\frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}} \\
&\geq \varsigma^2\frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}}.
\end{aligned} \tag{F.6}$$

in which I omitted the argument in $N(k)$ for ease of exposition. If $\max_m \frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}} = \frac{\mu^\omega_{N1}}{\mu^{\omega'}_{N1}} > K$, Equation (F.6) implies that

$$\frac{\mu^\omega_{Nm'}}{\mu^{\omega'}_{Nm'}} \geq \varsigma^{2(m'-1)}\frac{\mu^\omega_{N1}}{\mu^{\omega'}_{N1}} > \varsigma^{2(m'-1)}K.$$

As there must exist some m' with $\frac{\mu^\omega_{Nm'}}{\mu^{\omega'}_{Nm'}} \leq 1$, there must exist at least $\frac{\log K}{-2\log\varsigma}+1$ memory states with $\frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}} \geq 1$. Repeating the same analysis for other $\omega'' \in \hat{\Omega}(k)\setminus\{\omega\}$ implies that if $\max_m \frac{\mu^\omega_{Nm}}{\mu^{\omega''}_{Nm}} > K$ for all $\omega'' \in \hat{\Omega}(k)\setminus\{\omega'\}$, there must exist at $\frac{\log K}{-2\log\varsigma}+1$ memory states with a likelihood ratio $\min_{\omega'\in\hat{\Omega}(k)} \frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}} \geq 1$.

With the same argument, if $\max_m \frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}} > K$ for all $\omega \in \hat{\Omega}(k)$ and all $\omega' \in \hat{\Omega}(k)\setminus\{\omega\}$, there must exists $\frac{\log K}{-2\log\varsigma}+1$ memory states with a likelihood ratio $\min_{\omega'\in\hat{\Omega}(k)} \frac{\mu^\omega_{Nm}}{\mu^{\omega'}_{Nm}} \geq 1$ for all $\omega \in \hat{\Omega}(k)$. Therefore, $\frac{M}{|\hat{\Omega}(k)|} \geq \frac{\log K}{-2\log\varsigma}+1$. This implies that $\frac{M}{|\hat{N}|}$ goes to ∞ as K goes to ∞, which contradicts the fact that $\lim_{k\to\infty}\frac{M(N)}{|\hat{\Omega}(k)|} = \lim_{k\to\infty}\frac{M(k)}{N(k)} < \infty$ in big worlds.

Now I analysis the case of reducible automatons. Denote the recurrent communicating classes as $\mathscr{R}_{k1},\cdots,\mathscr{R}_{kr}$, and the set of transient memory states as $\mathscr{R}_{k0}$. The analysis above

applies in the cases where there is only one recurrent communicating class or where the initial memory state is in one of the recurrent communicating classes.

Now consider the case where $r > 1$, i.e., there are more than one recurrent communicating class, and the initial memory state denoted by i is in $\mathscr{R}_{k0}$. I first compute the probability of absorption by $\mathscr{R}_{kj}$ under state ω, denoted by $\mathscr{P}_k^{\omega}(\mathscr{R}_{kj})$. Consider a new transition rule $\mathscr{T}_k'$ where all transitions from $m \in \mathscr{R}_{k0}$ to another $m' \in \mathscr{R}_{k0}$ are the same as before. However, $\mathscr{T}_k'$ differs from $\mathscr{T}_k$ that all transitions from $m \in \mathscr{R}_{k0}$ to $m' \notin \mathscr{R}_{k0}$ are changed to transition from m to i. Given such a transition rule $\mathscr{T}_k'$, obviously only memory states in $\mathscr{R}_{k0}$ are reachable. Denote $\mu_{Nm}^{0\omega}$ as the stationary distribution of this new transition rule $\mathscr{T}_k'$.

As is known in the theory of Markov chain (see Appendix 2 of Hellman (1969)), $\mathscr{P}_k^{\omega}(\mathscr{R}_{kj})$ is given by:

$$\mathscr{P}_k^{\omega}(\mathscr{R}_{kj}) = \sum_{m \in \mathscr{R}_{k0}} \mu_{Nm}^{0\omega} \sum_{m' \in \mathscr{R}_{kj}} q_{mm'}^{\omega}$$

Also denote $\mu_{Nm}^{j\omega}$ as the stationary distribution within the recurrent communicating class $\mathscr{R}_{kj}$ given $\mathscr{T}_k$, i.e., $\mu_{Nm}^{j\omega} = \frac{\mu_{Nm}^{\omega}}{\sum_{m \in \mathscr{R}_{kj}} \mu_{Nm}^{\omega}}$, we have for $m \in \mathscr{R}_{kj}$

$$\begin{aligned}
\frac{\mu_{Nm}^{\omega}}{\mu_{Nm}^{\omega'}} =& \frac{\mathscr{P}_k^{\omega}(\mathscr{R}_{kj})}{\mathscr{P}_k^{\omega'}(\mathscr{R}_{kj})} \times \frac{\mu_{Nm}^{j\omega}}{\mu_{Nm}^{j\omega'}} \\
=& \frac{\sum_{m \in \mathscr{R}_{k0}} \mu_{Nm}^{0\omega} \sum_{m' \in \mathscr{R}_{kj}} q_{mm'}^{\omega}}{\sum_{m \in \mathscr{R}_{k0}} \mu_{Nm}^{0\omega'} \sum_{m' \in \mathscr{R}_{kj}} q_{mm'}^{\omega'}} \times \frac{\mu_{Nm}^{j\omega}}{\mu_{Nm}^{j\omega'}} \\
\leq& \varsigma^{-2} \frac{\sum_{m \in \mathscr{R}_{k0}} \mu_{Nm}^{0\omega}}{\sum_{m \in \mathscr{R}_{k0}} \mu_{Nm}^{0\omega'}} \times \frac{\mu_{Nm}^{j\omega}}{\mu_{Nm}^{j\omega'}} \\
\leq& \varsigma^{-2} \max_{m \in \mathscr{R}_{k0}} \frac{\mu_{Nm}^{0\omega}}{\mu_{Nm}^{0\omega}} \times \frac{\mu_{Nm}^{j\omega}}{\mu_{Nm}^{j\omega'}}.
\end{aligned}$$

Thus, if $\max_m \frac{\mu_{Nm}^{\omega}}{\mu_{Nm}^{\omega'}} > K$, we must have

$$\varsigma^{-2} \max_{m \in \mathscr{R}_0} \frac{\mu_{Nm}^{0\omega}}{\mu_{Nm}^{0\omega}} \times \max_{j \in \{1,2,\cdots,r\}} \max_{m \in \mathscr{R}_j} \frac{\mu_{Nm}^{j\omega}}{\mu_{Nm}^{j\omega'}} > K$$

Thus, we must have either

$$\begin{aligned}
\max_{m \in \mathscr{R}_0} \frac{\mu_{Nm}^{0\omega}}{\mu_{Nm}^{0\omega}} > \varsigma\sqrt{K}, \text{ or;} \\
\max_{j \in \{1,2,\cdots,r\}} \max_{m \in \mathscr{R}_j} \frac{\mu_{Nm}^{j\omega}}{\mu_{Nm}^{j\omega'}} > \varsigma\sqrt{K}.
\end{aligned}$$

Then the result follows similar arguments in the case of irreducible automatons. □

F.6 Proof of Proposition C.3

Proof. Suppose to the contrary that there exists a sequence of subsets of states $\tilde{\Omega}_k$ where $\lim_{k\to\infty} \frac{|\tilde{\Omega}_k|}{N(k)} > 0$, and that for all $\omega \in \tilde{\Omega}_k$, the DM takes action ω with some strictly positive probability in some state $\omega' \in \{1, \cdots, N(k)\}$ at the limit where $k \to \infty$ (where ω' could be different for different ω). Formally, this implies that for all $\omega \in \tilde{\Omega}_k$

$$\lim_{k\to\infty} \sum_{m\in M_k^\omega} \mu_{km}^{\omega'} \geq \xi$$

for some ω' and $\xi > 0$. As M_k^ω must be finite for almost all ω (measured in fraction), it implies that there exists some $\tilde{\Omega'}_k$ where $\lim_{k\to\infty} \frac{|\tilde{\Omega'}_k|}{N(k)} > 0$, and that for all $\omega \in \tilde{\Omega'}_k$ there exists some $\omega' \in \{1, \cdots, N(k)\}$,

$$\lim_{k\to\infty} \mu_{km}^{\omega'} \geq \xi' > 0 \text{ for some } m \in M_k^\omega.$$

Now denote the set of ω', that is, the set of states of the world where the DM picks some action $\omega \in \tilde{\Omega'}_k$ with strictly positive probability, as $\tilde{\Omega}''_k$. First, we must have $\lim_{k\to\infty} \frac{|\tilde{\Omega''}_k|}{N(k)} > 0$. Otherwise, there must exist some $\omega' \in \tilde{\Omega}''_k$ such that there are infinitely many memory states with probability bigger than ξ' which contradicts the fact that $\lim_{k\to\infty} \sum_{m=1}^{M} \mu_{km}^{\omega'} = 1$. With similar arguments, for almost all $\omega \in \tilde{\Omega'}_k$, there must exist some $m \in M_k^\omega$ such that $\lim_{k\to\infty} \mu_{km}^{\omega'} \geq \xi'$ for some $\omega' \in \tilde{\Omega}''_k$ but $\lim_{k\to\infty} \mu_{km}^{\omega''} = 0$ for almost all (measured in fraction) states $\omega'' \in \tilde{\Omega'}_k \setminus \{\omega'\}$. This implies that there exists some sequence of subsets of states $\tilde{\tilde{\Omega}}_k$ where $\lim_{N\to\infty} \frac{|\tilde{\tilde{\Omega}}_k|}{N} > 0$ such that

$$\lim_{k\to\infty} \max_m \frac{\mu_{km}^{\omega'}}{\mu_{km}^{\omega''}} = \infty$$

for all $\omega' \in \tilde{\tilde{\Omega}}_k$ and all $\omega'' \in \tilde{\tilde{\Omega}}_k \setminus \{\omega'\}$. However, this is shown to be impossible in the proof of Proposition C.2. The result thus follows.

□

F.7 Proof of Corollary C.3

Proof. Consider an example where $\lim_{N\to\infty} p_N^\omega = 0$ for all ω and $u_N(\omega', \omega) = 0$ for all ω and $\omega' \neq \omega$. As shown in Proposition C.3, all sequences of updating mechanisms must ignore almost all actions when k goes to infinity. Thus, $\lim_{k\to\infty} L_k^* = \lim_{k\to\infty} \sum_\omega u_{N(k)}(\omega, \omega) p_{N(k)}^\omega$ and all updating mechanisms are ϵ-optimal for all $\epsilon \geq 0$. Thus, if individual A adopts an updating

mechanism with $d_k(m) = 1$ for all m and k, while individual B adopts an updating mechanism with $d_k(m) = 2$ for all m and all k such that $N(k) \geq 2$, then they must disagree with each other under all ω for all big enough k such that $N(k) \geq 2$. □

F.8 Proof of Proposition E.1

Proof. The proof follows closely the proof of Proposition 1. For individual B, fixing the state ω, in the stationary probability distribution, we have at the two extreme memory states in branch ω',

$$\mu^{\omega}_{\omega'\lambda}\left[\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(G^{\omega''})+\gamma\right]=\mu^{\omega}_{\omega'(\lambda-1)}\left[(1-\gamma)F^{\omega}(G^{\omega'})+\frac{\gamma}{2}\right]$$

$$\mu^{\omega}_{\omega'(\lambda-1)}=\mu^{\omega}_{\omega'\lambda}\left[\frac{(1-\gamma)F^{\omega}(G^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(G^{\omega''})+\gamma}\right]^{-1}$$

for all ω'. Similarly, at memory state $\omega'(\lambda-1)$,

$$\mu^{\omega}_{\omega'\lambda}\left[\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\gamma\right]+\mu^{\omega}_{\omega'(\lambda-2)}\left[(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}\right]=\mu^{\omega}_{\omega'(\lambda-1)}\left[\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+(1-\gamma)F^{\omega}(S^{\omega'})+\gamma\right]$$

$$\mu^{\omega}_{\omega'(\lambda-2)}\left[(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}\right]=\mu^{\omega}_{\omega'(\lambda-1)}\left[\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}\right]$$

$$\mu^{\omega}_{\omega'(\lambda-2)}=\mu^{\omega}_{\omega'(\lambda-1)}\left[\frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}}\right]^{-1}$$

Repeating the same procedures implies that for all $k=1,\cdots,\lambda-1$

$$\mu^{\omega}_{\omega'k}=\mu^{\omega}_{\omega'\lambda}\left[\frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\gamma}\right]^{-1}\left[\frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}}\right]^{-(\lambda-k-1)} \tag{F.7}$$

and

$$\mu^{\omega}_{0}=\mu^{\omega}_{\omega'\lambda}\left[\frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\gamma}\right]^{-1}\left[\frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}}\right]^{-(\lambda-2)}\left[\frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{N}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega'}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}}\right] \tag{F.8}$$

Note that for all $\gamma\in(0,1)$

$$\frac{(1-\gamma)F^{\omega}(S^{\omega})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}}>1.$$

$$\frac{(1-\gamma)F^{\omega}(S^{\omega})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}} > \frac{(1-\gamma)F^{\omega}(S^{\omega'})+\frac{\gamma}{2}}{\frac{1-\gamma}{\delta}\sum_{\omega''\neq\omega}F^{\omega}(S^{\omega''})+\frac{\gamma}{2}}.$$

Then the result follows from the same steps in the proof of Proposition 1. □

References

Hellman, Martin E and Thomas M Cover (1970) "Learning with finite memory," *The Annals of Mathematical Statistics*, Vol. 41, No. 3, pp. 765–782.

Hellman, Martin Edward (1969) *Learning with finite memory* Ph.D. dissertation, Standford University.